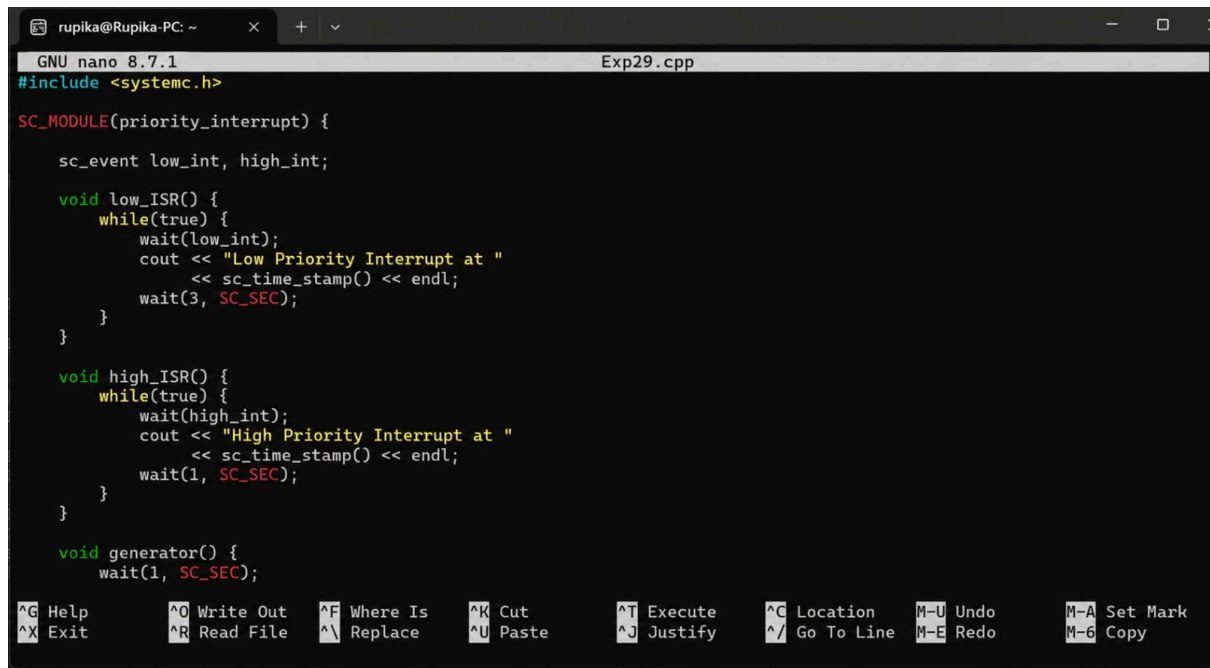


EXP:29



The screenshot shows a terminal window with the GNU nano 8.7.1 editor open. The file being edited is 'Exp29.cpp'. The code defines a module for priority interrupts, including variables for low and high interrupts, and functions for their respective interrupt service routines (ISR) and a generator function. The low_ISR function prints 'Low Priority Interrupt at ' followed by the current time stamp and waits for 3 seconds. The high_ISR function prints 'High Priority Interrupt at ' followed by the current time stamp and waits for 1 second. The generator function waits for 1 second before calling the low_ISR function.

```
GNU nano 8.7.1 Exp29.cpp
#include <systemc.h>

SC_MODULE(priority_interrupt) {

    sc_event low_int, high_int;

    void low_ISR() {
        while(true) {
            wait(low_int);
            cout << "Low Priority Interrupt at "
                << sc_time_stamp() << endl;
            wait(3, SC_SEC);
        }
    }

    void high_ISR() {
        while(true) {
            wait(high_int);
            cout << "High Priority Interrupt at "
                << sc_time_stamp() << endl;
            wait(1, SC_SEC);
        }
    }

    void generator() {
        wait(1, SC_SEC);
    }
}
```

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy

```
rupika@rupika-PC:~$ nano Exp29.cpp
rupika@rupika-PC:~$ g++ -o Exp29 Exp29.cpp -lsystemc
rupika@rupika-PC:~$ ./Exp29
```

```
SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
Copyright (c) 1996-2025 by all Contributors,
ALL RIGHTS RESERVED
```

```
Low Priority Interrupt at 1 s
High Priority Interrupt at 2 s
```

```
rupika@rupika-PC:~$ |
```